WHEN THE QUOTIENT RULE APPLIES

The limit of a quotient is the quotient of the limits, provided the denominator's limit is nonzero. When it is zero the rule fails, and the numerator's limit decides what happens instead.

QUOTIENT RULE

$$\lim_{x \to a} \frac{f(x)}{g(x)}$$

split by the limits L of f and M of g

$L = \lim_{x \to a} f(x)$ and $M = \lim_{x \to a} g(x)$

CASE 1

$M \neq 0$

The rule applies

the limit of the quotient is the quotient of the limits

$$\frac{L}{M}$$

CASE 2

$M = 0,\ L \neq 0$

Typically infinite

analyze the sign from each side to decide between $+\infty$ and $-\infty$

CASE 3

$M = 0,\ L = 0$

Indeterminate form $0/0$

carries no information until the expression is rewritten; see further techniques

TAKEAWAY

Check M first. Only when $M = 0$ does L matter: a nonzero L means the quotient grows without bound, and $L = 0$ leaves the indeterminate form $0/0$.